Total No. of Questions : 5]

SEAT No. :

PC1670

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T.Y.B.Sc. (Cyber & Digital Science)

CDS - 351 : DIGITAL FORENSICS - I

(2020 Pattern) (Semester- V)

Time : 3 Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *All questions are compulsory.*
2) *Figures to the right indicate full marks.*
3) *Draw diagrams wherever necessary.*

Q1) Attempt any Ten of the following: **[10×1=10]**

a) Define Cyber Crime.
b) Which are the two types of memory types?
c) Define Digital evidence.
d) Write down any two Computing Environment.
e) What is GUI?
f) Define Digital Forensics.
g) Name any two File Systems.
h) Define Search Warrant.
i) Define DHCP.
j) Define Forensic Workstations.
k) Name any two primary storage.
l) Define Mobile Forensics.

Q2) Attempt any Five of the following: **[5×2=10]**

a) What is Law enforcement?
b) What is digital file metadata?
c) Explain challenges of Acquiring Digital Evidence.
d) What are the firewalls?
e) What are Validation Protocols?
f) What are network Forensics?

Q3) Attempt any Four of the following: **[4×5=20]**

a) Explain Digital Forensics and its goals.

b) Solve the following.

i) Convert the binary number 111011001 into a decimal number.

ii) Convert 111102 into a decimal number system.

c) Explain Chain of Custody in detail.

d) Explain the different types of Computing Environment.

e) Explain Order of Volatility.

Q4) Attempt any Four of the following: **[4×5=20]**

a) What is Cybercrime Attack Mode? How computers are used in cybercrimes?

b) Explain Different types of Computer Storage.

c) Explain in detail - Digital Forensics Examination Process.

d) Explain incident response process.

e) Explain Digital Forensics Hardware Tools.

Q5) Attempt any One of the following: **[1×10=10]**

a) Explain Different types of digital forensics. **[5]**

b) Differentiate between Digital Forensics vs. Other Computing Domain. **[3]**

c) Explain Computer Encoding Character Schema. **[2]**

OR

a) Explain the Duty of First Responder Tasks. **[5]**

b) Write note on: Capturing Network Traffic. **[3]**

c) Explain Digital Evidence types. **[2]**

Total No. of Questions : 5]

SEAT No. :

PC1671

[Total No. of Pages : 3

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T.Y. B.Sc. (Cyber & Digital Science)

CDS - 352 : CYBER THREAT INTELLIGENCE

(2020 Pattern) (Semester - V)

Time : 3 Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *All questions are compulsory.*
2) *Figures to the right indicate full marks.*
3) *Draw diagrams wherever necessary.*

Q1) Attempt any Ten of the following: **[10×1 = 10]**

a) Name one tool used for analyzing cyber threat data.

b) What is threat intelligence?

c) Define cyber incident information-sharing.

d) What does APT stand for in cybersecurity?

e) What is standardization in information sharing?

f) What is the primary goal of intelligence communities in cybersecurity?

g) Name one infrastructure component essential for CTI communities.

h) Define data breach in the context of cybersecurity?

i) What is the main purpose of situational awareness models in cybersecurity?

j) Define information leakage in cybersecurity.

k) What is one challenge associated with sharing threat information?

l) What is the main goal of national threat intelligence (TI)?

Q2) Attempt any five of the following: **[5×2 = 10]**

a) Identify two challenges organizations face in threat information sharing.

b) How do tooling and infrastructure support CTI communities?

c) What is the primary purpose of monitoring raw data in cybersecurity?

d) How does the legal and regulatory landscape affect cybersecurity information sharing?

e) What are two primary responsibilities of cybersecurity centers?

f) List two benefits of having a well-structured CTI community.

Q3) Attempt any four of the following: **[4×5 = 20]**

a) Elaborate on each step of the cyber kill chain, highlighting the importance of each phase in preventing cyber-attacks.

b) Compare and contrast two methods of analyzing raw monitoring data to derive cybersecurity insights.

c) What are the responsibilities and tasks of cybersecurity centers in ensuring national cybersecurity, explain with examples.

d) Explain the role of infrastructure in supporting the growth and collaboration within a CTI community.

e) Explain the term disproportionate mitigation measures and provide one example.

Q4) Attempt any four of the following: **[4×5 = 20]**

a) Explain the benefits and challenges of cross-organizational threat information sharing, providing examples for each.

b) How does efficient cooperation and coordination help in achieving effective information sharing in cybersecurity?

c) Explain the relationship between cybercrime and cybersecurity, including how they influence each other in the modern digital landscape.

d) What are the advantages and limitations of using raw monitoring data for cyber threat analysis?

e) What are the potential legal barriers to data transfer in cybersecurity and how organizations can navigate them?

Q5) Attempt any one of the following: **[1×10 = 10]**

a) Explain the roles, responsibilities, and processes within a National TI Framework and how they contribute to its success. **[5]**

b) Describe one technology used to implement a National TI Framework and its importance. **[3]**

c) What are the key challenges in deriving insights from raw monitoring data? **[2]**

OR

a) How can organizations balance the need to share security-relevant information while protecting personal data and maintaining anonymity? **[5]**

b) What role does the EU Cybersecurity Legal Framework play in cross-border data transfer? **[3]**

c) Explain the role of anonymization in sharing security-relevant information. **[2]**

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Total No. of Questions : 5]

SEAT No. :

PC1672

[6333]-53

[Total No. of Pages : 2

T.Y.B.Sc (Cyber & Digital Science)
CDS - 353 : INFORMATION SECURITY POLICY AND AUDIT
(2020 Pattern) (Semester-V)

Time : 3 Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***Figures to the right indicate full marks.***
3) ***Draw diagrams wherever necessary.***

Q1) Attempt any TEN of the following : **[10 x 1=10]**

a) What is IT governance?

b) What is Information Security?

c) Define HMAC.

d) Define Network security.

e) What is Cryptography.

f) What is the role of a Forensic Investigator?

g) What is an Audit report?

h) What is the role of a public key in asymmetric encryption?

i) Define Cloud Computing.

j) List the different threats that affects the information Security?

k) What is the purpose of logging controls?

Q2) Attempt any FIVE of the following : **[5×2=10]**

a) What is Operating system security? Name two common methods used in operating system security.

b) What is the main difference between Encryption and Hashing?

c) What is Computer forensics? Name two common tools used in computer forensics.

d) What is COBIT? What is the main purpose of COBIT?

e) Define Computer-Assisted Audit Tools (CAATs). Name two common types of CAATs.

f) Explain the following terms

i) User Security Policy

ii) System Security Policy

Q3) Attempt any FOUR of the following : **[4×5=20]**

a) What is Contingency plan? Explain its Stages.

b) Write down the steps of information security audit reports.

c) Explain ISO/IEC 27001 standard for information security.

d) Explain the importance of incident reporting in maintaining organizational security. How incident reporting process help in detection and mitigation of security threats?

e) Define Domain. Explain the following terms

i) Active Directory

ii) Group Policy

iii) Organizational Unit (OU)

iv) Directory permissions

Q4) Attempt any FOUR of the following : **[4×5=20]**

a) What is the difference between Symmetric and Asymmetric key Cryptography?

b) What are the Frameworks & Standards for Cyber Security Domains?

c) Explain Network Attacks and also explain hosted and web based attacks.

d) Short note on Log monitoring

e) Explain the concept of risk management in information security.

Q5) Attempt any One of the following : **[1×10=10]**

a) Explain ISMS briefly. **[5]**

b) What is Physical Security ? Explain the types of Physical Security. **[3]**

c) Define IPSec **[2]**

OR

a) What strategies do you use to identify' areas of vulnerability within an organization's IT infrastructure? **[5]**

b) What is Digital Signature ? Explain its working in detail. **[3]**

c) Define Disaster recovery. **[2]**

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Total No. of Questions : 5]

SEAT No. :

PC-1673

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T.Y. B.Sc. (Cyber & Digital Science)

PROFESSIONAL ELECTIVE - I

CDS-357A Mobile Forensics

(2020 Pattern) (Semester - V)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***Figures to the right indicate full marks***
3) ***Drawdiagramswherevernecessary.***

Q1) **Attempt any eight of the following :** **[8 × 1 = 8]**

a) Define Mobile Stations

b) What is a SIM?

c) What is Micro Read?

d) NAND memory is made up of multiple _____.

e) Which Cellebrite UFED solutions can perform Physical acquisition of mobile devices?

f) Which acquisition method can be used to retrieve data from severely damaged devices?

g) Who are the main users of the ParabeniRecovery Stick?

h) Name the tool that can decode and analyze Call Data Records (CDRs)?

I) Which is the proprietary file system developed by Apple?

j) What is logical acquisition?

P.T.O.

Q2) Attempt any Four of the following : **[4 × 2 = 8]**

a) Enlist Acquisition Methods

b) Enlist the categories of mobile operating systems

c) Name two types of data that the iRecovery Stick can retrieve from an ioS device. And explain how

d) Justify True or False "In the evidence acquisition phase, the first task is to identify the mobile device make and model. This information includes the device manufacturer and the device specific model."

e) Enlist Micro Systemation AB (MSAB)

Q3) Attempt anyTwo of the following : **[2 × 4 = 8]**

a) What can affect Artefacts extraction from a mobile device?

b) Write a note on phases of Mobile Forensics Process.

c) What is Oxygen Forensics?

Q4) Attempt anyTwo of the following : **[2 × 4 = 8]**

a) Write a note on Examination and Analysis phase of mobile forensics process

b) What is the primary purpose of the ParabeniRecovery Stick?

c) Explain why Bypassing the security controls on mobile devices is not an easy task

Q5) Attempt any One of the following : **[1 × 3 = 3]**

a) Write a note on SIM security

OR

b) What is Physical Acquisition?

Total No. of Questions : 5]

SEAT No. :

PC-1674

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T.Y. B.Sc.
CYBER & DIGITAL SCIENCE
CDS-357B: Cloud Security
(2020 Pattern) (Semester - V) (Professional Elective - I)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***Figures to the right indicate full marks.***
3) ***Draw diagrams wherever necessary.***

Q1) **Attempt any eight of the following :** **[8 × 1 = 8]**

a) What are the main service models of cloud computing?

b) What is data classification?

c) What are the cloud assets?

d) What are the key components of an IAM system.

e) What is Finding Leaks?

f) What is Compute Resource? Write any one example.

g) What is utility computing?

h) What do you mean by Infrastructure as a Service (IaaS)?

i) What role do Single Sign-On (SSO) solutions play in IAM?

j) Why is revalidation important for maintaining security?

Q2) **Attempt any Four of the following** : **[4 × 2 = 8]**

a) What are the characteristics of confidential data?

b) Differentiate the private cloud and public cloud.

c) What are the three data classification levels?

d) Define the CREATE operation in the context of IAM. What types of entities can be created?

e) How can tooling leaks impact data security?

f) Write a short note on network assets.

Q3) **Attempt any Two of the following :** **[2 × 4 = 8]**

a) What are the three pillars of data security?

b) What is Community cloud? Explain in details.

c) What is biometric authentication? Discuss the different types of biometric methods.

Q4) **Attempt any Two of the following :** **[2 × 4 = 8]**

a) Which are the different data identification services available?

b) What are the main types of access control models?

c) Why Multi-Factor Authentication (MFA) is important in IAM?

Q5) **Attempt any One of the following :** **[1 × 3 = 3]**

a) Describe the Cloud Computing Architecture.

b) Explain the lifecycle for Identity and Access management.

OR

b) What are the four pillars of data asset management?

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